



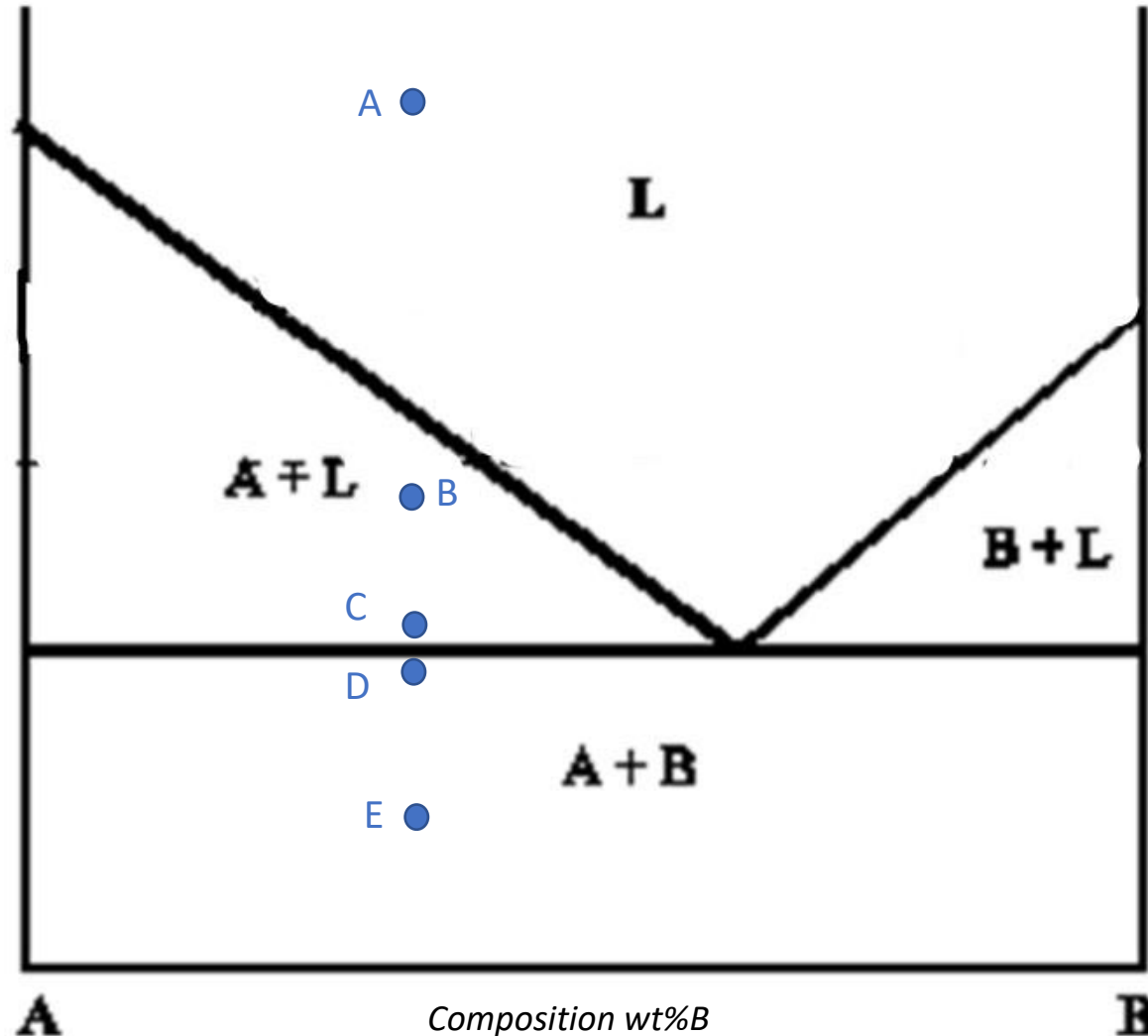
STM Exercise 6

Phase Diagrams

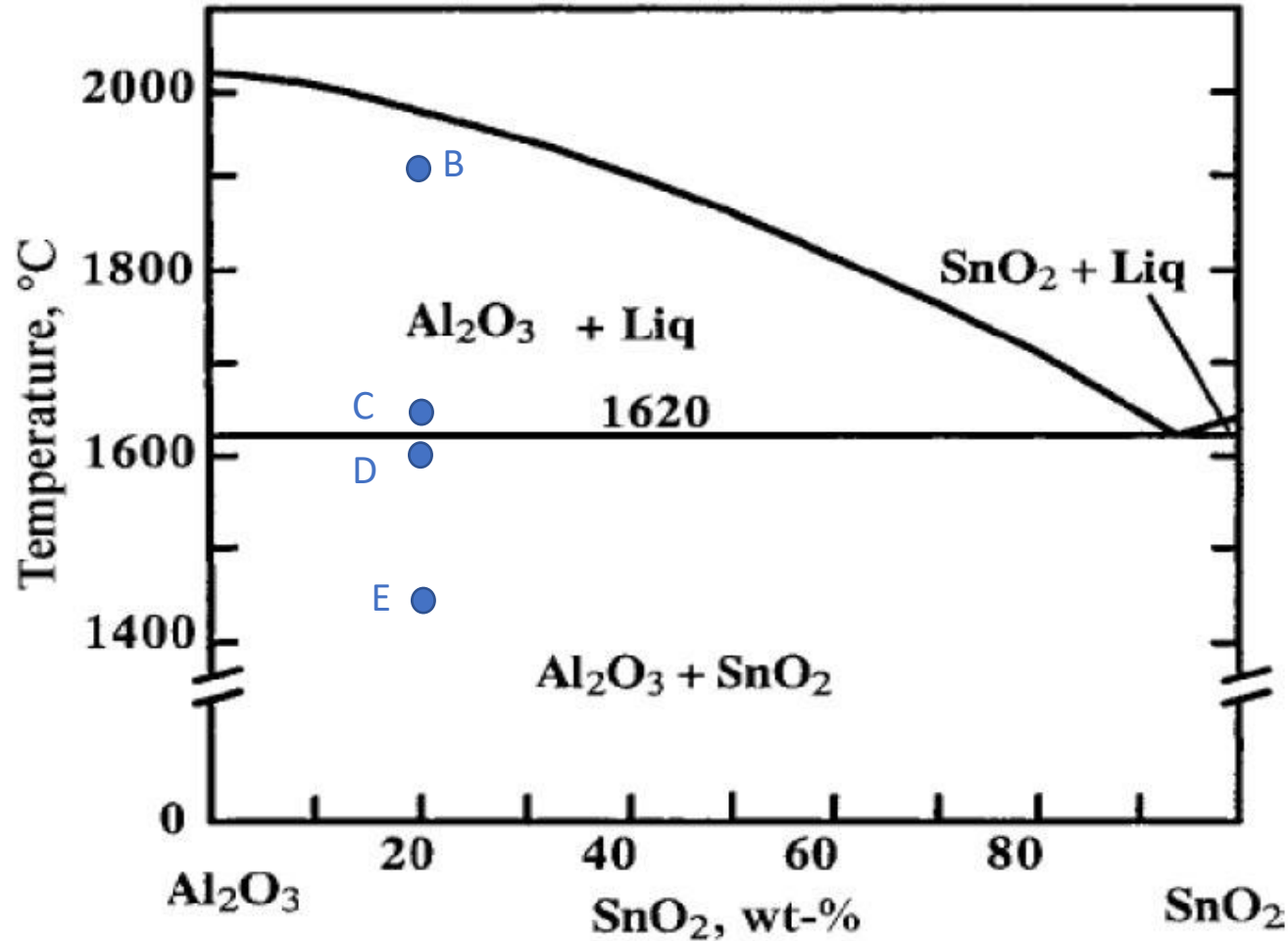
(no solubility in the solid state)

(partial solubility in the solid state)

Exercise 1: determine the phase present, their composition and relative amounts in the point **A**, **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.

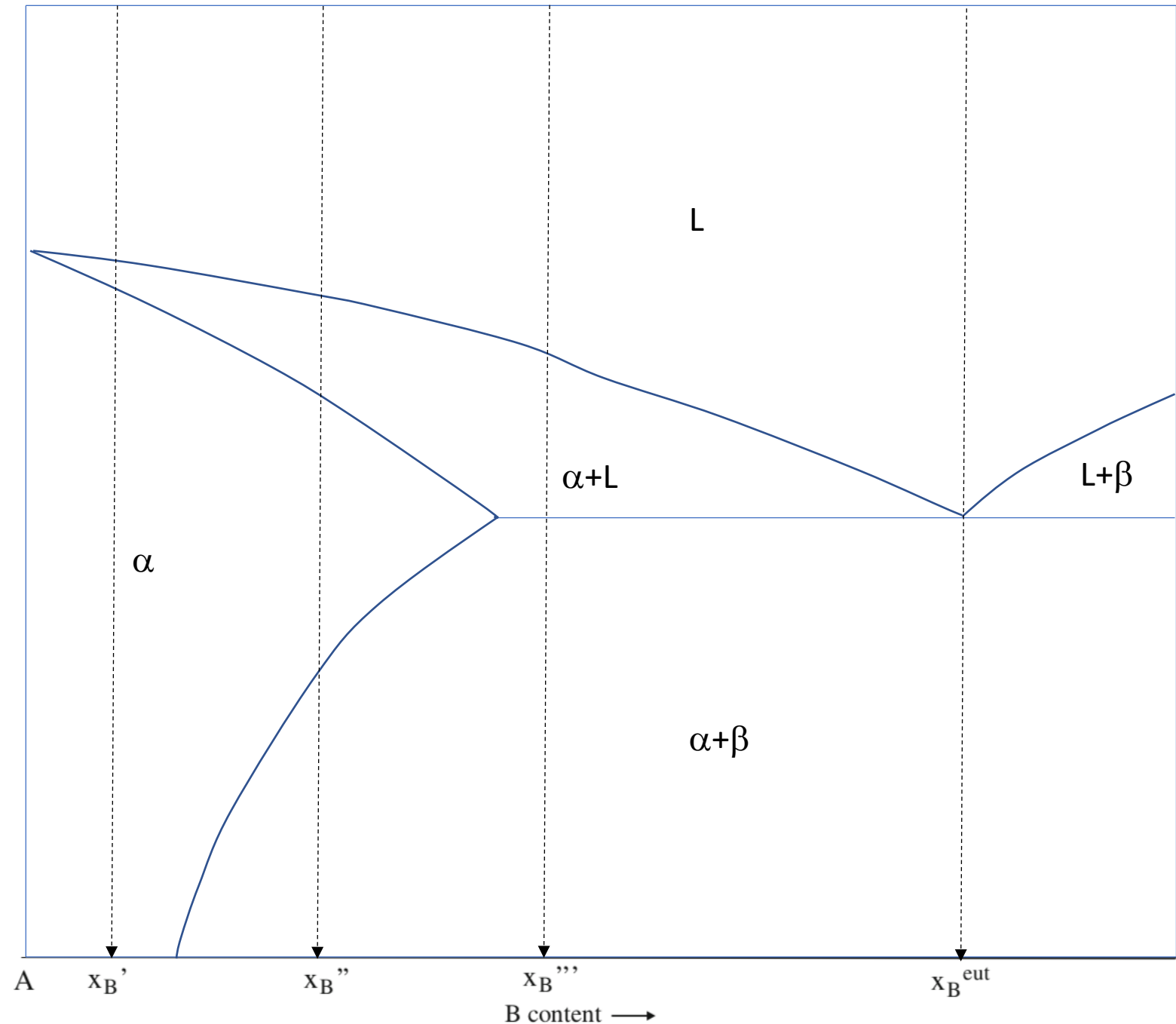


Exercise 2: determine the phase present, their composition and relative amounts in the points **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



Exercise 3: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **partial solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

Exercise 4: describe the solidification of alloys with compositions x_B' , x_B'' , x_B''' and x_B^{eut} by drawing schematic microstructures in significant points.



Exercise 5: determine the phase present, their composition and relative amounts in the points **A**, **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.

